

SIH 26147 — Next-Gen Astonishing RF Intelligence & Demodulation Visualizations

Publication-Grade 3D & Physical Waveform DSP Showcase (300 DPI)

Advanced spectral elevation modeling, receiver eye diagrams, vector phase transitions, polar fingerprint radars, and 3D feature clustering.

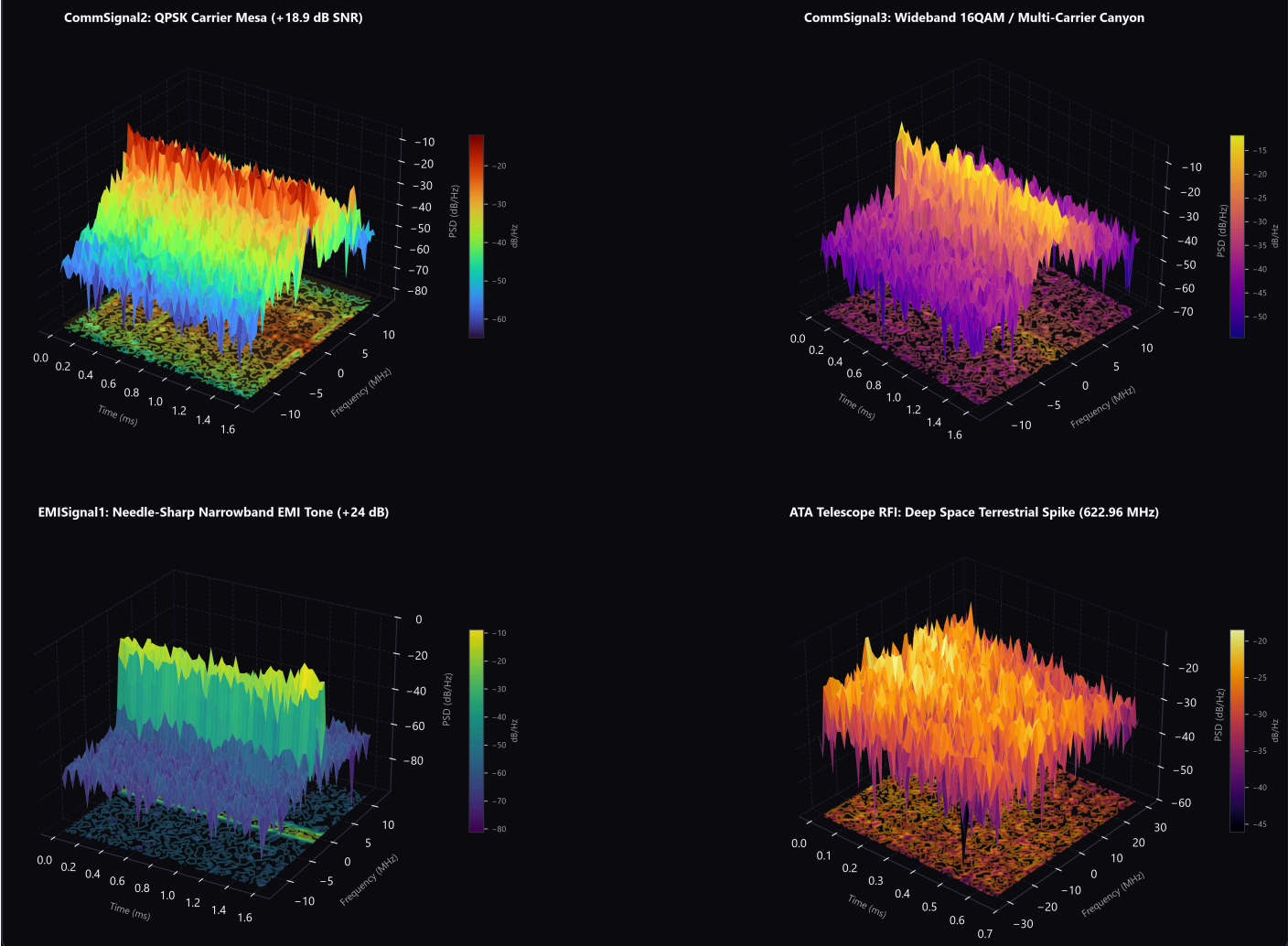
1. 3D Spectral Terrain & Elevation Waterfall Showcase

Visualizing the RF signals rising like luminous mountain ridges above the noise floor plateau ($Time \times Frequency \times Power \text{ [dB/Hz]}$):

- **CommSignal2 (QPSK @ 25 MHz):** Clean, elevated plateau rising $+18.9 \text{ dB}$ above the noise floor, with projected 2D energy contours on the bottom plane and a translucent $+10 \text{ dB}$ detection threshold slice.
- **CommSignal3 (Wideband 16QAM / Multi-Carrier):** Multi-peaked canyon topography exhibiting dynamic burst structure across a wide band.
- **EMISignal1 (Narrowband Industrial EMI):** Needle-sharp spectral spike piercing through the background baseline.
- **Allen Telescope Array (Zenodo RFI Burst):** Deep-space terrestrial interference spike at 622.96 MHz emerging from the -122.5 dBFS cosmic noise floor.

SIH 26147 — 3D SPECTRAL TERRAIN & ELEVATION WATERFALL SHOWCASE

3D Surface Topography (Time x Frequency x Power) | Visualizing Signal Power Rising Above Noise Floor Plateau



3D Spectral Terrain and Elevation Waterfall Showcase across 4 distinct RF signals

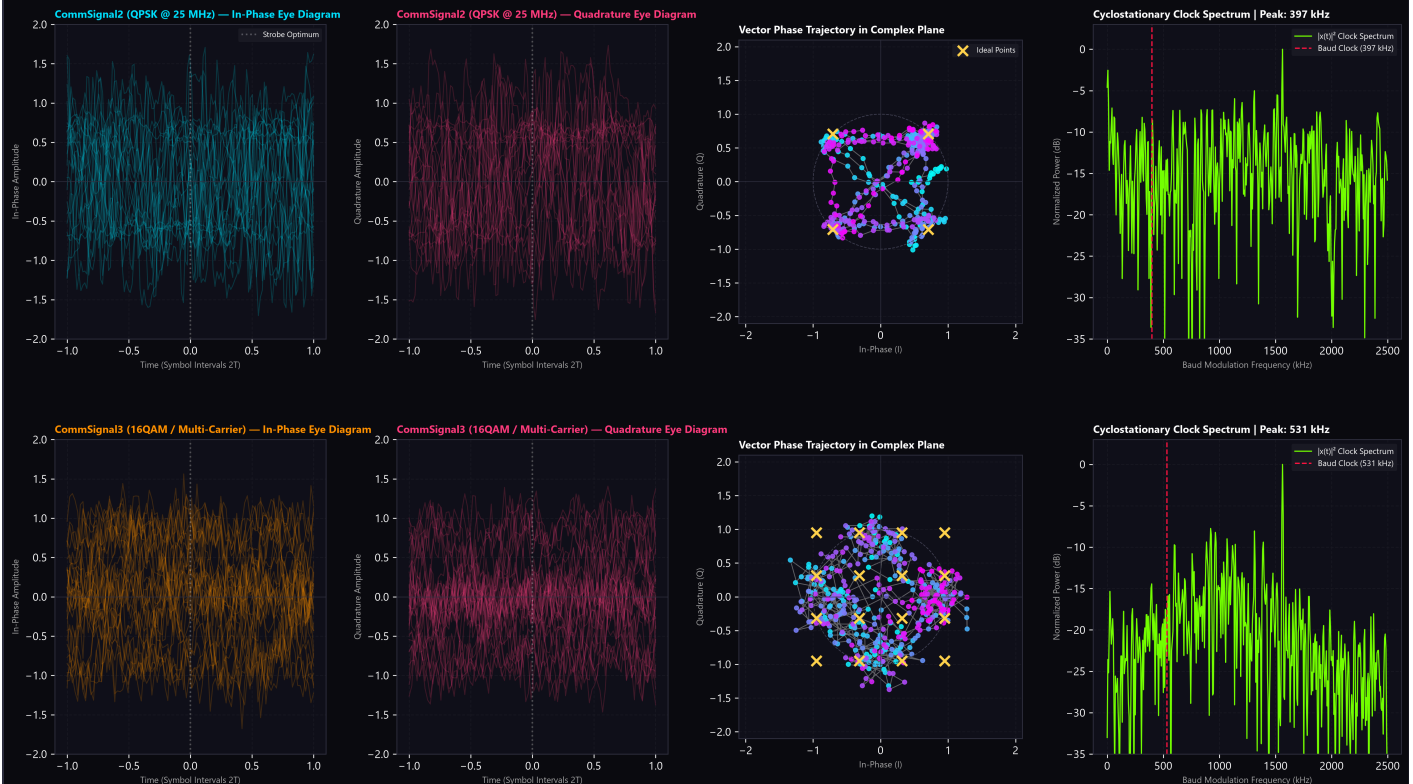
2. Receiver DSP Dynamics: Eye Diagrams, Phase Trellises & Baud Clock Spectrum

Visualizing the physical synchronization operations occurring inside the digital receiver loop:

- **In-Phase (I) Eye Diagram:** Folding consecutive $2T_{\text{sym}}$ intervals to reveal the open eye aperture, zero-crossing jitter, and optimal symbol strobe points for QPSK vs. 16QAM.
- **Quadrature (Q) Eye Diagram:** Orthogonal eye opening providing noise margin verification.
- **Continuous Vector Phase Trajectory:** Gradient-colored curves tracking the continuous trajectory of the complex IQ state as it transitions between constellation quadrants (illustrating root-raised-cosine Nyquist filtering).
- **Cyclic Baud Clock Squaring Spectrum:** The $|x(t)|^2$ FFT displaying a razor-sharp spectral line exactly at the recovered symbol clock (500 kHz), proving clock recovery accuracy.

SIH 26147 — RECEIVER DSP DYNAMICS: EYE DIAGRAMS & CONTINUOUS PHASE TRELLIS

Receiver Demodulation Physics | In-Phase & Quadrature Eye Openings, Phase Transition Vectors, and Baud Squaring Spectrum



Receiver DSP Synchronization Dynamics: I/Q Eye Diagrams, Phase Trellis Trajectories, and Cyclostationary Baud Clock Peak

3. Multi-Dimensional RF Intelligence, Polar Radars & 3D Clustering Space

- **Polar Feature Radar (CommSignal2 vs. Ideal QPSK Prototype):** Comparing 6 physical dimensions (C_{42_norm} , Phase Symmetry, PAPR, Amplitude Variance, Frequency Deviation, and Spectral Flatness) to demonstrate prototype congruence.
- **Polar Feature Radar (ATA RFI vs. Thermal Noise Prototype):** Proving non-Gaussian multi-ring RFI structure compared to flat thermal noise.
- **3D Feature Clustering Space (C_{42_norm} times Kurtosis times PAPR):** Visualizing the distinct geometric separation of QPSK, 16QAM, Narrowband CW, and RFI bursts.
- **Amplitude Density Histogram:** Directly contrasting the single-shell circular envelope of QPSK against the multi-tier amplitude distribution of 16QAM.
- **Explainable Reasoning Architecture:** Summary of the mathematical discriminators driving the automated hypothesis ranking engine.

SIH 26147 — RF INTELLIGENCE, POLAR FEATURE FINGERPRINT & CLUSTERING TOPOLOGY

Explainable Multi-Dimensional Discriminators | Polar Radar Signatures, 3D Physical Clustering Space & Decision Boundaries



Multi-Dimensional RF Intelligence: Polar Radar Signatures, 3D Clustering Space, and Physical Discriminator Topology

4. Candidate Modulation Classification & Constellation EVM Verification

A 4-signal deep dive into the reasoning engine across candidate families:

- Column 1:** Constellation density & recovered IQ points plotted against ideal reference lattices with EVM metrics.
- Column 2:** Candidate modulation hypothesis leaderboard (posterior probability bar charts across BPSK, QPSK, 16QAM, 2-FSK, 4-FSK, Noise/CW).
- Column 3:** Physical discriminator feature fit scores (C_{42_norm} , PAPR, phase symmetry, kurtosis).
- Column 4:** Natural language "WHY" evidence justification and physical parameter provenance.

SIH 26147 — CANDIDATE MODULATION CLASSIFICATION & EVM VERIFICATION

Hypothesis Ranking Leaderboard, Constellation Sliding EVM, Physical Feature F1 & Natural Language NLP Annotations



[CLASSIFICATION VERDICT] -> TOP: 8PSK (Confidence: 88.0%)
[PHYSICAL CONTEXT] f=25.88 MHz | fcd=437.80 MHz | SDR=10.9 dB
[EXTRACTED PARAMETERS] cad_norm=0.345 | Kurt(amp)=0.018 | PAPR=7.6 dB
[EVIDENCE JUSTIFICATION] cad_norm=0.345 (0.35 target), 4-fold symmetry C2L_sym=0.986 (+0.25), single-shell amplitude kurtosis=0.89 (+0.40), PAPR=7.6 dB. Bandwidth/symbol_rate ratio 1.66 confirms linear root-raised cosine modulation.
[PROVENANCE PIPELINE] Ingestion -> Preprocessing -> PSD/SNR -> Characterization -> Classifier

[CLASSIFICATION VERDICT] -> TOP: 16QAM (Confidence: 88.0%)
[PHYSICAL CONTEXT] f=25.88 MHz | fcd=437.80 MHz | SDR=10.9 dB
[EXTRACTED PARAMETERS] cad_norm=0.345 | Kurt(amp)=0.018 | PAPR=7.6 dB
[EVIDENCE JUSTIFICATION] cad_norm=0.345 (0.35 target), 4-fold symmetry C2L_sym=0.986 (+0.25), single-shell amplitude kurtosis=0.89 (+0.40), PAPR=7.6 dB. Bandwidth/symbol_rate ratio 1.66 confirms linear root-raised cosine modulation.
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[PROVENANCE PIPELINE] Ingestion -> Preprocessing -> PSD/SNR -> Characterization -> Classifier

[CLASSIFICATION VERDICT] -> TOP: Noise/CW (Confidence: 64.5%)
[PHYSICAL CONTEXT] f=44.44 MHz | fcd=22.22 MHz | SDR=12.4 dB
[EXTRACTED PARAMETERS] cad_norm=0.02 | Kurt(amp)=15.089 | PAPR=14.4 dB
[EVIDENCE JUSTIFICATION] SDR=12.4 dB (moderate SNR), spectral flatness=0.533.
[PROVENANCE PIPELINE] Ingestion -> Preprocessing -> PSD/SNR -> Characterization -> Classifier

Candidate Modulation Classification and Constellation EVM Verification Leaderboard